Tianyi Xu

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Research Interests

I develop efficient and adaptive methods for AI agents, spanning multi-agent systems, test-time training and adaptation, self-evolution, and data-centric AI, where capability, reliability, and compute efficiency must be jointly optimized.

Education

Oregon State University — *Sept 2026 – Present*
Ph.D. in Computer Science
Advisor: Huazheng Wang

University of Wisconsin-Madison — *Sept 2021 – Dec 2025*
B.S. in Computer Science, Data Science, and Mathematics; GPA: 3.918/4.000

Publications

** denotes equal contribution.*

Published

[1] **OPUS: Towards Efficient and Principled Data Selection in Large Language Model Pre-training in Every Iteration** [arXiv] [Blog] — ICML 2026; DATA-FM @ ICLR 2026
Shaobo Wang*, Xuan Ouyang*, Tianyi Xu*, Yuzheng Hu, Jialin Liu, Guo Chen, Tianyu Zhang, Junhao Zheng, Yubo Ma, Kexin Yang, Xingzhang Ren, Dayiheng Liu, Linfeng Zhang
ICML Spotlight (top 2.2%); Top #1 Paper of the Day at Hugging Face Daily

[2] **Stem rot affects the structure of rhizosphere microbiome in berseem clover (*Trifolium alexandrinum*)** [DOI] — Rhizosphere, 2025
Salma Mukhtar, Zain Ahmad, Noor Khan, Tianyi Xu, Dalaq Aiysha

Under Review

[3] **EvoPool: Evolutionary Programmatic Annotation for Label-Efficient Specialized Supervision** [arXiv] — Under Review
Tianyi Xu, Yaolun Zhang, Xuan Ouyang, Huazheng Wang

[4] **EVOCHAMBER: Test-Time Co-evolution of Multi-Agent System at Individual, Team, and Population Scales** [arXiv] — Under Review
Yaolun Zhang*, Tianyi Xu*, Shengyu Dai, Zhenwen Shao, Qingyun Wu, Huazheng Wang

[5] **Generating the Unheard: Phylogeny-Guided Latent Generation for Ancestral Sound Reconstruction** — Under Review
Tianyi Xu, Shrinaath Narasimhan, Evan Gorstein, Santiago Perea, Yunyi Shen, Claudia Solís-Lemus

[6] **MAST: Label-Efficient, Robust, and Generalizable Sound Detection for Biodiversity Monitoring via Masked Audio Pretraining and Self-Training** — Under Review
Tianyi Xu, Daniel L. Pimentel-Alarcón, Zuzana Buřivalová, Claudia Solís-Lemus

[7] **SITA: Learning Speaker-Invariant and Tone-Aware Speech Representations for Low-Resource Tonal Languages** [arXiv] — Under Review

Tianyi Xu*, Xuan Ouyang*, Binwei Yao, Shoua Xiong, Sara Misurelli, Maichou Lor, Junjie Hu

[8] **Combined effects of methyl bromide and soil amendments on soil bacterial and fungal communities in turfgrass** [Preprint] — Under Review

Tianyi Xu*, Salma Mukhtar*, Evan Gorstein, Claudia Solís-Lemus, Ming Yi Chou, Paul Koch

Talks / Presentations

Friday at 4 Seminar – Invited Talk — Dec 2025

"Self-Supervised Sound Detection with AudioMAE for Robust, Label-Efficient Biodiversity Monitoring"

Department of Plant Pathology, University of Wisconsin–Madison

Undergraduate Research Symposium – Poster Presentation — Apr 2024, Apr 2025

26th & 27th Annual Undergraduate Symposium, University of Wisconsin–Madison
"Biodiversity Detection Using Acoustic Signals with Deep Learning"
Presented as part of the Holstrom Environmental Fellowship requirement (2025).

Abstract (2025, p. 164), Abstract (2024, p. 139)

Awards

ICML 2026 Spotlight (Top 2.2%), Seoul — 2026

Phi Beta Kappa (PBK), Inducted — 2025

Holstrom Environmental Research Fellowship — 2024

Awarded $3,000 + $1,000 (advisor) for research on acoustic event detection in tropical rainforests using deep learning.

UW–Madison Summer Scholarship — 2024

Awarded $1,000 for Summer 2024 studies in recognition of outstanding academic achievement.

Dean's List (Six Semesters) — 2021–2024

Recognized for maintaining a GPA above 3.85 in Fall 2021, Spring 2022, Fall 2022, Fall 2023, Spring 2024, and Fall 2024.

Research Experiences

Research Intern, Epic Lab (SJTU) — *Remote*
Advisor: Linfeng Zhang, Mentor: Shaobo Wang — *Aug 2025 – Feb 2026*

- Co-developed **OPUS**: an **optimizer-aware**, efficient, validation-guided **data selection** method for LLM pre-training with **Adam** and **Muon** optimizers.
- Built the **distributed pre-training and continued pre-training stack** for Llama3, Qwen3, and GPT-2 using **Megatron** with FSDP and TP on multi-GPU, and **vLLM**-based evaluation; **+2 to 3 pts** avg on downstream tasks.

Research Assistant — *Madison, WI*
Advisor: Junjie Hu — *July 2025 – Present*

- Led **SITA**, addressing **tone-collapse** in multilingual self-supervised speech models for **low-resource tonal languages** by adapting wav2vec2 and XLS-R into **tone-aware, speaker-invariant** representations.
- Designed a multitask training recipe combining contrastive learning, ASR CTC auxiliary loss, and **knowledge distillation**, implemented in **fairseq** with distributed multi-GPU training; achieved **0.95 to 0.99** Top-1 and Top-5 cross-speaker retrieval while preserving tone distinctions.

Research Assistant — *Madison, WI*
Advisor: Claudia Solís-Lemus — *June 2025 – Present*

- Led **MAST**, a label-efficient and shift-robust framework for **time-frequency biodiversity sound detec-**

tion combining masked audio pretraining, audio-aware ViT adapters, **box-level contrastive learning**, and **iterative self-training**; provided theoretical analysis casting self-training as **distributionally robust optimization** with monotonic improvement guarantees, achieving **+0.22 mAP** and **+0.24 F1** over the strongest baseline under cross-site shift on rainforest soundscapes.

- Led **Generating the Unheard**, the first generative framework for **ancestral sound reconstruction** of bird vocalizations, combining VAE latent encoding, a learned **tree-metric trait projection** aligned with phylogenetic distances, **Brownian ancestral inference**, and a closed-form **anchored inverse lift**; proved row-orthonormality and Brownian motion posterior optimality, validated on Tyrannidae and Paridae clades.